



UNIVERSITI TEKNOLOGI MALAYSIA

Technical Report

Enterprise Systems Design & Modelling (SECP3744)

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Executive Summary

This technical report shows how blockchain technology improves the transparency and trust in the enterprise supply chain. Traditional enterprise systems like ERP and SCM always struggle with fragmented data, limited traceability and also lack of trust between organizations. Blockchain which is a decentralized and also unchangeable record keeping system can tackle these issues by securing and verifying all transactions across multiple stakeholders.

This paper reviews the potential of blockchain to improve the traceability, reduce scam and also streamline operation in industries such as food, energy and pharmaceuticals. Using real time data sharing and smart contracts, blockchain offers more efficient collaboration between multiple organizations in the business. However, there are also challenges in the blockchain such as scalability, privacy and interoperability.

The analysis concludes that to successfully implement the blockchain, pilot projects, integrations with existing enterprise systems and a good framework is needed. Continued research should focus on standardization, hybrid architectures, and cross-industry collaboration

1.0 Introduction

In this modern era, supply chains have become more complex where it involves multiple parties, cross-border transactions and large amounts of data exchanged among independent organizations. Transparency, traceability and trust are still a common barrier with which global companies must take note to maintain ethical sourcing, sustainability and agreement based on international regulations (Saber et al., 2019). Blockchain technology, popularized by cryptocurrencies in its early days has the potential to effectively help manage with these challenges due to secure decentralized and tamper-proof data management (Treiblmaier, 2018)

In the area of enterprise systems, blockchain offers a new means of implementing interorganizational collaboration with its inclusion of key business processes such as logistics, production, etc. By integrating blockchain in enterprise resource planning (ERP) or supply chain management (SCM) databases, organizations can obtain real-time visibility of product flows and valid records of product movement and ownership (Kshetri, 2021). This integration will help better decision making, improve accountability, and minimise the risk related to fraud, counterfeiting, and data tampering (Cole et al., 2019).

The objective of this report is to analyze how blockchain technology can enhance the transparency and trustworthiness in the corporate supply chain. It explores how blockchain functions, looks up the research and case studies, what are the benefits and drawbacks, and the potential of blockchain in the future. The discussion will focus on the impact of blockchain on supply chain operations through unalterable data, use of smart contract, and compatibility with already existing enterprise systems. In the end, this paper aims to deliver useful insights to companies that want to use blockchain to create supply chains with the highest level of transparency, efficiency and trustworthiness.

2.0 Background / Literature Review

As companies have developed their supply chain globally, their supply chains have become more complex. These new circumstances have created challenges for them—data is siloed, it is harder to detect fraud, and engagement with allied partners can feel like herding cats. Researchers have argued that traditional enterprise systems are sufficient to manage operations within a single enterprise, but have a hard time creating transparency and trust between enterprises (Sabeti et al., 2019). Blockchain may potentially disrupt this status quo, as it can provide a consistent record that may or may not, be distributed, immutable, and verifiable (Treiblmaier, 2018). Think of blockchain as a shared digital ledger, where information is recorded on many computers creating simultaneous transactions. Once the data is recorded it is difficult to change in any significant way without a majority of the people or all of the people agreeing to the change. This creates a system where the experts we trust are not needed, this is what is referred to as "trustless", trust being the absence of the need for a mutually trusted intermediary (Kshetri, 2021). Researchers have been exploring how to incorporate the use of blockchain into enterprise systems for some time now, especially in relation to supply chain and ERP systems, In an attempt to facilitate visibility, traceability of products, and accountability (Cole et al., 2019; Kouhizadeh et al., 2020).

We are already seeing some very interesting practical examples. For example, IBM Food Trust uses blockchain to track food products from the farm to the store shelf, which enhances the ease with which food products can be validated for safe and authentic status (van Hoek, 2020). Pharmaceutical industry is also using blockchain in order to track the medications throughout the supply chain, while also complying with international regulations (Kouhizadeh et al., 2020). This shows how blockchain can tell where information is lacking by using the automated validation process while increasing the collaboration among partners.

Even though blockchain gives a lot of advantages, it's not come without difficulties. Researchers also find several challenges such as privacy issues, scalability and interoperability with existing enterprise systems remain significant barriers to widespread adoption (Sabeti et al., 2019). Furthermore, the lack of framework for the standardized blockchain creates uncertainty. Thus new research continues to solve this problem such as combining IOT or AI to achieve a more secure supply chain.

3.0 Analysis and Discussion

3.1 How the Technology Works

In its essentials, the Blockchain technology is a system of distributed ledgers, with every transaction (or data line) being bundled into a block that is systematically connected to the previous one, a system of which is widely distributed between participants in the blockchain network (also called nodes). This is what makes this architecture so that after a block is appended, it is not easy to manipulate without the agreement of the network - this provides immutability and auditable transparency.

A supply-chain example with blockchain In a supply-chain case where one integrates enterprise systems (e.g. enterprise resource planning (ERP) or supply-chain management (SCM) systems), blockchain is used as a common data store: individual organisations store events (e.g. shipping, receipt, quality-check)

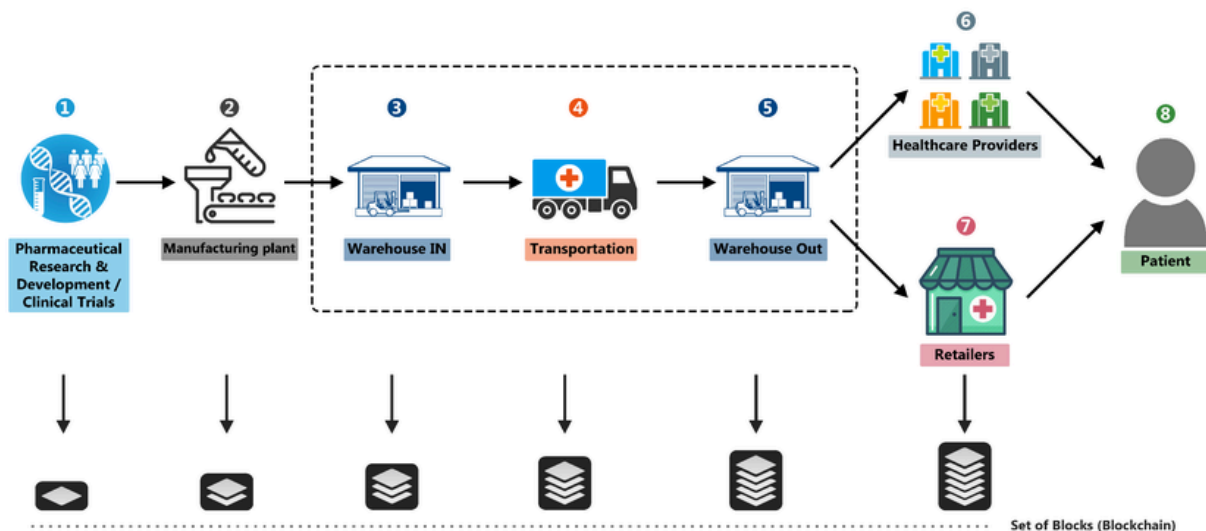
in the ledger, smart contracts process automatically (e.g. payment release upon delivery) and everyone has a single version of the truth.

For Example, the ledger can be permissioned (enterprise-scale) such that only trusted actors (manufacturers, logistics providers, retailers) have write or path validation rights; however, the data is stored in a resistant version between them. Smart contracts inside the blockchain are used to perform such automation tasks as ensuring a shipment inherited created conditions (temperature, origin, timestamp) are met and then proceeded with

3.2 Case Examples & Applications

A notable application is in the food-supply use-case: organisations use blockchain to track produce farm-to-shelf, to quickly issue a recall in case of an incident; and to ensure the authenticity of assets. A system named Vind was suggested to run provenance of software/hardware assets in a complicated supply chain of energy-delivery systems, such that produce provenance is recorded and audit logs provide assurance.

The other example of a robust enterprise is the unification of blockchain and ERP applications: a research has yielded new tendencies of decentralising ERP-based activities, automating procurement across organisations and managing data security in supply-chains through blockchain-ERP hybrids through an illustrative view:



3.3 Advantages and Disadvantages

Advantages

- **Transparency & Traceability:** Every movement or event is recorded, allowing stakeholders and even consumers to verify origin, pathway and legitimacy of goods.

- **Data Integrity & Trust:** Because records cannot easily be changed retroactively, dependence on trusted intermediaries can reduce; partners can rely on the shared ledger.
- **Automation:** Smart contracts enable self-executing agreements (e.g., automated payment when delivery confirmed), reducing delays and manual errors.
- **Improved Collaboration:** In multi-party supply-chains, blockchain fosters a shared infrastructure that aligns data across enterprises, reducing silos.

Disadvantages

- **Scalability and Performance:** Blockchain systems may face transaction throughput or latency issues, especially when many participants or high event rates are involved.
- **Integration with Legacy Systems:** Existing enterprise systems (ERP, WMS, MES) were not designed for blockchain-native interaction; middleware and adaptation are often required.
- **Privacy and Governance:** Sharing data across partners risks exposing sensitive information; balancing openness with confidentiality is non-trivial.
- **Adoption & Change Management:** Getting all supply-chain partners on board, aligning standards, and managing change is a major organisational challenge.

3.4 Challenges and Future Trends

Even with the promise, there are numerous hurdles, which blockchain will still have to face before it can become a default infrastructure in the supply chains of businesses. The key challenges are the measurability of interoperability between various blockchain platforms and enterprise systems, the creation of governance models of multi-party networks, and regulatory and data-sovereignty issues across borders. A research on ERP-blockchain integration has observed the existence of industry-specific modular, scalable blockchain architectures requirement paradigm

Emerging trends point to hybrid architectures combining blockchain with the Internet of Things (IoT) for real-time sensor data on goods (temperature, humidity), or with Artificial Intelligence (AI)/analytics for predictive supply-chain insights. For example, micro-service-based blockchain traceability systems have been proposed to handle high-volume data flows and adapt to changing business processes

4.0 Recommendations

For enterprises that intends to implement blockchain in their supply chain operations, there are several strategic and technical recommendations can greatly enhance effectiveness and sustainability :

1. Start with pilot projects

Companies should test blockchain in a small and controlled environment such as a product line or supplier network tracking. A pilot implementation will help locate difficulties in the actual integration, requirement for governance, and return on investment (ROI) potential before make it full scale deployment

2. Integrate with existing enterprise systems

Blockchain should not replace the current ERP or SCM systems, rather it should be a complement. API-based connectors can synchronize data between traditional databases and distributed ledger, thus allowing the information flow efficiently and avoiding data silos

3. Develop governance and data sharing protocol.

A well defined governance structure where it includes ownership, access privileges etc is essential. In addition, models were formed as consortium (rather than a single entity) would govern the private blockchain. This governing model will be the best for transparency while still protecting sensitive data.

5.0 Conclusion

Blockchain technology introduces a new approach for enhancing transparency and trust within an organization's supply chain. The technology provides a tamper-evident and decentralized ledger, promoting organizations to safely record and share data among multiple untrusted stakeholders. Linking this system with enterprise system such as ERP and SCM can help organizations to improve the traceability of data, the integrity of data, and operational efficiency

Despite the promise of blockchain technology for trustworthy, open supply chains etc, it is inhibited by real world constraints of scalability, data privacy and the synchronization across multiple systems. Even though adoption of blockchain technology has not been used vastly in many contexts, improvement in combining blockchain technology with existing tech, automated contract, and IoT are steadily moving things forward.

Over a long term, companies that can successfully manage to implement blockchain technology with clear rules and collaborative processes will be more likely to build and maintain an open and reliable worldwide supply chain into the future . Further studies should explore the connection between systems, regulations, and measurable sustainability outcomes for future implementations.

6.0 References

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7.0 Appendix / Logbook

link: <https://github.com/users/mukhritz/projects/19/views/1>

